

Recentring Academics — Syllabus Redesign: ENG-SCI 53 Quantitative Physiology as a Foundation for Bioengineering

- **Course:** ENG-SCI 53 (Engineering Sciences 53 / "ES 53"; the bioengineering foundation course), Fall 2025. Head Instructor: Dr. Linsey Moyer.
- **Source syllabus:** [inputs/syllabus/ENG-SCI 53 - 2025 Fall \(122339\).pdf](#).
- **Date:** 2026-06-10

ENG-SCI 53 builds active learning into its structure: students "take an active part in the learning process," a weekly three-hour lab has them use "basic medical and physiologic instrumentation to record real-time data sets," and faculty and alumni panels are woven into the term, so the engaged-classroom pillar rests on supervised, hands-on work rather than on exhortation. With respect to grade differentiation, the course places 40% of the grade on three in-class, seated exams (75 minutes, one page of notes) that are both verifiable and high-ceiling, and it states a criterion-referenced intent: "We want your grade to reflect what you get out of the course, not what you brought into it." The differentiation gap is in where the rest of the weight sits: the lab reports (18%) and the final project (14%) are group submissions graded one-per-team, so individual mastery is not separately verified, and the problem sets (18%) are take-home, which leaves the exams carrying most of the verifiable differentiation. That spine holds up only if the exams reserve marks for the problems that only the strongest students clear.

1. Audit

Category	Current Syllabus	Guidance	Assessment
Attendance & participation	"Because of the interactive nature of this course, we highly encourage attendance. Participation will be necessary for you to completely achieve the course goals." "Attendance at these [faculty/alumni] meetings will be part of the participation grade." Participation is 5%.	Clear, communicated attendance and participation expectations on a stated standard (course policies)	Participation is tied to attendance at class and at the faculty/alumni panels, and reinforced by the in-class quizzes. Attendance is "highly encourage[d]" rather than required, and the syllabus does not state a quality standard for participation beyond presence.
In-class device use	Not addressed (the syllabus has a "Privacy Policy" on recording, but no policy on student device use).	Limit or prohibit internet-enabled devices except when directed (course policies)	A no-device policy for the lecture meetings is recommended; the labs already direct device and instrument use by activity.

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Generative-AI policy	"P-Sets: We specifically forbid the use of ChatGPT or any other generative artificial intelligence (AI) tools for any problem sets unless coding in MATLAB is required... Coding for Labs: We encourage the use of the MATLAB AI Playground, ChatGPT-Edu and Harvard-based generative AI to help you learn to code and complete the post-lab analysis for all of the labs. Violations of this policy will be considered academic misconduct."	Clear, learning-objective-aligned AI policy made real by assignment design (course policies; designing assignments)	A thoughtful component-specific policy: AI is off-limits for the quantitative reasoning on problem sets but welcomed for lab coding, tied explicitly to learning goals. The problem-set ban is take-home and not fully enforceable; differentiation appropriately rests on the supervised exams. The policy names academic misconduct but does not reference the Honor Council.
Grade Weighting	Problem sets 18% (one dropped), lab reports 18%, in-class quizzes 5%, exams 40% (lowest exam only 10%), final project presentation 14%, participation 5%.	Enough of the grade must rest on work that can differentiate on content/skill	The 40% on seated exams is verifiable, high-ceiling work that can differentiate on mastery. The 18% lab reports and 14% project are group submissions, and the 18% problem sets are take-home, so half the grade rests on help-exposed or team-based work where individual mastery is harder to read.

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Grade Differentiation	In-class quizzes are 10-minute "comprehension checks"; exams "cover lecture material, laboratory material, and problems similar to those found in your problem sets."	Assessments must span enough difficulty to differentiate at the top, not top out for most strong students; past ~15% at the maximum a component is not resolving the top (discriminating-at-the-top)	The 5% in-class quizzes are short comprehension checks that top out for prepared students. Differentiation rests on the three exams, which can resolve the top if a portion of each sits above the difficulty of the practiced problem sets; the syllabus does not specify whether it does. Group-graded lab reports compress individual differences by construction.
Assessment Variety	Quantitative problem sets, group laboratory reports with MATLAB analysis, short in-class quizzes, three seated exams, and a team design project pitched in person.	Mastery tested across varied formats	Mastery is tested across quantitative, experimental, computational, written, and oral/design formats, so the grade is not tied to one mode of assessment.
Verifiability/AI-Resilience	Exams held in class (75 minutes, one page of notes); in-class quizzes in class; problem sets take-home ("each student's problem set must represent his/her/their own work," AI forbidden except MATLAB); lab reports "one report is submitted per group" of three, AI encouraged for coding; final project a team-of-four pitch "in person during the scheduled final exam period."	Work should be reliable evidence of the student's own unaided thinking; seated exams broaden the distribution (Harvard report , §III)	About 45% of the grade (exams 40% + quizzes 5%) is produced under supervision and is structurally verifiable. The 18% problem sets are take-home; the 18% lab reports and 14% project are team submissions whose individual contribution the syllabus requires but does not separately grade. The in-person project pitch is observable but team-based.
Cumulative assessment	Three midterm exams across the term (the last in the final week); the final project synthesizes "one specific clinical issue" and is pitched during the final exam period;	End-of-term assessment should be cumulative (designing assignments ; capstone)	The final project is a cumulative design capstone in spirit, but it is a team task, not an individually verifiable cumulative assessment,

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	there is no separate cumulative final exam.		and the exams are organized by unit rather than cumulatively.
Grading rigor & range	"We want your grade to reflect what you get out of the course, not what you brought into it." "Tentative Grading Breakdown." No explicit A–F standard stated.	Grade to explicit criterion-referenced standards (Sadler 2005); A reserved for "extraordinary distinction" (proposal)	Criterion-referenced in spirit. The syllabus does not state how the components convert to letter grades or what distinguishes excellent from satisfactory work on an exam, a lab report, or the project.
Feedback & consistency	One teaching fellow plus seven course assistants and lab staff grade problem sets, lab reports, and exams.	Norm grades across graders; rater training and calibration improve reliability (rubrics ; Jonsson & Svingby 2007)	With many graders across problem sets, lab reports, and exam free-response, consistency is a live risk. The syllabus mentions a lab-report guide and template but does not describe grader calibration on shared samples; without it, lab-report and free-response grading carry inconsistency risk.

2. Recommended revisions

- **Add a device policy for the lecture meetings.**
 - *Currently:* none (the syllabus addresses recording under "Privacy Policy" but not student device use).
 - *Revise to:* "Internet-enabled devices are not to be used during lecture except when an instructor directs their use for an activity." The labs already direct instrument and device use, so this only adds a lecture-meeting expectation ([course policies](#)).
- **Make the lab reports and final project carry individually verifiable, differentiating evidence.**
 - *Currently:* "one report is submitted per group" of three; the final project is graded as a "teams of four" pitch.
 - *Revise to:* attach a graded individual component to each — for example, an individually written and individually graded section of the lab report (the syllabus already requires each student to contribute to the writing and MATLAB analysis), and a short individual question-and-answer at the project pitch in which each team member explains their own contribution and its physiological reasoning, so the grade reflects individual mastery rather than only the team's product ([full-range-task-design](#); [discriminating-at-the-top](#)).
- **Confirm the exams reserve marks for items only the strongest clear.**

- *Currently:* exams cover "problems similar to those found in your problem sets."
- *Revise to:* ensure a portion of each exam sits above the difficulty of the practiced problem sets (transfer or modeling problems applied to an unfamiliar organ-system scenario) and check the fraction of students at full marks after grading ([discriminating-at-the-top](#)).
- **Add an end-of-term cumulative element.**
 - *Currently:* three unit exams and a team final project, with no cumulative individual assessment.
 - *Revise to:* make a portion of the last exam, or an individual component of the final project, cumulative across the organ systems studied, so end-of-term work integrates the whole course rather than the most recent unit ([designing assignments](#); [capstone](#)).
- **Publish standards and calibrate grading across the teaching team.**
 - *Currently:* a lab-report template is provided, but with one TF and seven CAs no calibration process is described.
 - *Revise to:* write analytic rubrics for the lab reports and exam free-response whose top band is left open for genuine insight rather than a checklist, distribute annotated exemplars, and hold a calibration session where graders score shared samples before grades are set ([rubrics](#); [rubrics-differentiation](#); Jonsson & Svingby 2007).

3. A tailored generative-AI policy

The current policy is already component-specific and well-aligned to the course's goals; it need only be stated against which components are verifiable and which are help-exposed, with an enforcement route. Suggested syllabus language:

Generative AI in ENG-SCI 53.

- On problem sets, you may not use generative-AI tools for the quantitative physiology reasoning; AI is permitted only for MATLAB coding and debugging where a problem requires it. The reasoning must be your own.
- On labs, you are encouraged to use approved generative-AI tools (the MATLAB AI Playground, ChatGPT-Edu, and Harvard-based tools) to learn to code and complete the post-lab MATLAB analysis. Cite where you used them, and remain responsible for the correctness of what you submit.
- The in-class quizzes and the seated exams are completed without AI; they are where your individual mastery is assessed.

If we have reason to believe submitted work used AI outside these terms, we may ask you to discuss it with us and, if warranted, refer the matter to the Honor Council.